Signed-graph union: independent source review

Native post-find union branch

Choose orientation; read fixed pairs; attach child; transport or compare anchor

↓

Checked whole-state invariant

Every vertex equation and every stored anchor survive both orientations

↓

Independent controls

Target-offset reversal; signed grandchild; retained anchors; both-fixed answer

↓

Zero-constraint boundary

Valid state proves constant0; invalid-state model none does not model panic

↓

Composition still separate

Preceding finds; forest/rank/size; native words/arrays; compiled execution